

Krishnaa Vadivel | Computational Physics PhD | First-Principles Materials | HPC | Scientific ML

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Summary

Computational physics researcher combining first-principles condensed-matter theory, many-body modeling, and production scientific software. Experience spans distributed HPC on Frontier and Perlmutter, CUDA/ROCm acceleration, semiconductor simulation, and ML methods for materials and molecular systems.

Research Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Develop first-principles and many-body approaches for self-trapped excitons and excitonic polarons in real materials, connecting DFT, DFPT, GW/BSE-style workflows, and reduced effective models.
- Build distributed simulation workflows in Python and C++ for leadership-class HPC systems, using MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier, Perlmutter, and other national lab clusters.
- Formulate coherent-state and path-integral viewpoints for exciton-phonon dynamics and investigate ML accelerators, including equivariant graph neural networks, for faster materials modeling.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS research in 2024, 2025, and 2026.

Industry Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Implemented embedded and sensor-side functionality across Lattice FPGA, PIC32, dsPIC, STM32, and C8051 platforms; supported circuit bring-up, oscilloscope-based debugging, and selective PCB updates in Altium.
- Contributed host-side tooling with C#, ASP-based services, and Microsoft SQL-backed workflows for data handling and engineering operations.

FindLight

Photonics Technical Writer

2018–2019

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Exciton-Phonon HPC Research: Distributed first-principles calculations for self-trapped excitons, many-body formulations, and scalable workflows for materials simulations.

Semiconductor Device Simulation: Numerical modeling of semiconductor equations, transport, and pn-junction-style device behavior for physics and EE audiences.

Equivariant Molecular ML: Graph neural network workflows for molecular-property prediction and ML-assisted scientific computing.

Agentic Scientific Automation: RAG-style tooling for reading documentation and research papers to generate structured computational-science inputs and workflows.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, SciPy, SymPy, matplotlib

ML: PyTorch, PyTorch Geometric, scikit-learn

Platforms: Linux, macOS, Windows, Docker, Docker Compose, Kubernetes, gcloud

Domains: First-principles materials modeling, many-body theory, semiconductor physics, numerical PDEs